

A more resilient operation

Northstar Manufacturing · 2026-10-01 · Base case

THE OBJECTIVE

Reduce avoidable operating costs while building a reliable data foundation.

Decision owner: Strategy team

36-month NPV	36-month ROI	Sustained payback
\$502,501	107.3%	18 months

Selected initiatives & delivery

INITIATIVE	OWNER	DELIVERY
Document intelligence Minutes per invoice	Finance operations	Months 1–2
Operational data foundation Asset records with complete history	Data team	Months 1–3
Predictive maintenance Unplanned downtime hours	Plant operations	Months 4–8

Months 0–12

Costs \$421,333
Benefits \$225,417
Period ROI -46.5%

Months 13–36

Costs \$194,000
Benefits \$1,050,000
Period ROI 441.2%

Monthly model in USD at 10.0% annual discount rate. Period ROI is not an independent investment return. Estimates depend on supplied assumptions; see the complete appendix.

Assumptions & evidence appendix

36-month NPV: \$502,501 | ROI: 107.3% | Sustained payback: Month 18

All selected initiatives have a feasible schedule.

Financial model

USD. 10.0% annual discount rate. Implementation costs at start; recurring costs and benefits after delivery; benefits ramp over 3 months. Scenario benefit multiplier 1, cost multiplier 1, delay 0 months. No inflation, tax, residual value or monetized soft benefits.

Months 0–12: costs \$421,333, benefits \$225,417, ROI -46.5%.

Months 13–36: costs \$194,000, benefits \$1,050,000, ROI 441.2%. Period ROI is not an independent investment return.

Implementation funding: months 1–3 \$180,000; months 4–12 \$450,000; months 13–36 \$300,000. Concurrent delivery capacity: 3 people. Industry: Manufacturing.

Portfolio and roadmap

- Document intelligence — delivery months 1–2; Fits the requested start, funding and capacity.
Problem: Manual invoice review slows payment and creates rework.. Owner: Finance operations; KPI: Minutes per invoice; risk: Low; prerequisites: none; external: none (not confirmed).
- Operational data foundation — delivery months 1–3; Fits the requested start, funding and capacity.
Problem: Fragmented maintenance records prevent reliable analysis.. Owner: Data team; KPI: Asset records with complete history; risk: Medium; prerequisites: none; external: none (not confirmed).
- Predictive maintenance — delivery months 4–8; Starts after its prerequisites finish.
Problem: Unplanned downtime interrupts production.. Owner: Plant operations; KPI: Unplanned downtime hours; risk: Medium; prerequisites: Operational data foundation; external: none (not confirmed).
- Supplier service assistant — Not selected
Problem: Routine supplier questions interrupt procurement.. Owner: Procurement; KPI: Hours spent on routine inquiries; risk: Medium; prerequisites: none; external: none (not confirmed).
- Demand forecasting — Not selected
Problem: Forecast error increases inventory carrying costs.. Owner: Supply chain; KPI: Weighted forecast error; risk: High; prerequisites: Operational data foundation; external: none (not confirmed).

Assumption appendix

- Document intelligence: implementation \$65,000; annual operating \$18,000; annual full-adoption benefit \$155,000; 1 delivery people for 2 months, requested start month 1. Impact 4/5; effort 2/5 (qualitative ratings).
- Operational data foundation: implementation \$95,000; annual operating \$24,000; annual full-adoption benefit \$40,000; 2 delivery people for 3 months, requested start month 1. Impact 3/5; effort 3/5 (qualitative ratings).
- Predictive maintenance: implementation \$210,000; annual operating \$55,000; annual full-adoption benefit \$330,000; 2 delivery people for 5 months, requested start month 4. Impact 5/5; effort 4/5 (qualitative ratings).
- Supplier service assistant: implementation \$85,000; annual operating \$28,000; annual full-adoption benefit \$125,000; 1 delivery people for 3 months, requested start month 4. Impact 3/5; effort 2/5 (qualitative ratings).
- Demand forecasting: implementation \$160,000; annual operating \$40,000; annual full-adoption benefit \$230,000; 2 delivery people for 4 months, requested start month 13. Impact 4/5; effort 4/5 (qualitative ratings).

Evidence ledger

- Document intelligence / Annual benefit: Synthetic demonstration fixture; Finance operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Document intelligence / Implementation cost: Synthetic demonstration fixture; Finance operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.

- Document intelligence / Annual operating cost: Synthetic demonstration fixture; Finance operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Operational data foundation / Annual benefit: Synthetic demonstration fixture; Data team; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Operational data foundation / Implementation cost: Synthetic demonstration fixture; Data team; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Operational data foundation / Annual operating cost: Synthetic demonstration fixture; Data team; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Predictive maintenance / Annual benefit: Synthetic demonstration fixture; Plant operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Predictive maintenance / Implementation cost: Synthetic demonstration fixture; Plant operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Predictive maintenance / Annual operating cost: Synthetic demonstration fixture; Plant operations; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Supplier service assistant / Annual benefit: Synthetic demonstration fixture; Procurement; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Supplier service assistant / Implementation cost: Synthetic demonstration fixture; Procurement; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Supplier service assistant / Annual operating cost: Synthetic demonstration fixture; Procurement; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Demand forecasting / Annual benefit: Synthetic demonstration fixture; Supply chain; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Demand forecasting / Implementation cost: Synthetic demonstration fixture; Supply chain; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.
- Demand forecasting / Annual operating cost: Synthetic demonstration fixture; Supply chain; 2026-09-05; Unverified; Synthetic example. Illustrative estimate for exploring the product. Replace with verified operating evidence.

Decision history

No decisions recorded.

Limitations

Estimates are conditional on supplied assumptions. Capacity represents concurrent delivery people; operating staffing is not scheduled. Fixed earliest-fit scheduling is not a global scheduling optimum. Data remains in this browser unless exported or explicitly sent to the optional AI assistant. Generated by CanvasIQ.